Chengze Li

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RESEARCH EXPERIENCE

Shanghai AI Laboratory, *Research Intern - Agentic Multimodal AI* Shanghai, China. 05/2026-08/2026

- Led MedClaim-RL, an agentic multimodal medical discovery system on a private HNSCC cohort of 1,200 patients, aligning H&E WSIs, 3D CT/MRI volumes, pathology/radiology reports, EHR variables, and OS/PFS endpoints into patient-level evidence graphs for claim-level biomarker and recurrence hypothesis generation.
- Implemented a 6-agent medical claim pipeline that routes each patient through planning, retrieval, multimodal encoding, EHR extraction, statistical testing, and critique, aligning Qwen3-VL/InternVL3.5 report reasoning with UNI2/TITAN WSI features and nnU-Net/Swin-UNETR 3D imaging features to generate claim-specific Cox/XGBoost notebooks with FDR, calibration, and leakage checks.
- Optimized agent policies via SFT warm-start, DPO-style clinician preference learning, and KL-regularized offline RL over 20K logged tool-use rollouts, using a claim-level reward for external ΔC-index/AUROC, WSI-CT-text agreement, code pass rate, ECE/Brier calibration, and clinician preference minus leakage/site-overfit penalties; filtered 240 hypotheses to 42 claims through locked-site validation, TCGA-HNSC transfer, sham-endpoint controls, and supervised-only/retrieval-only/no-RL ablations with claim-level audit logs.

Manteia Technologies Co., Ltd. Milwaukee, WI

Research Scientist 04/2025-06/2025

Research Scientist Intern 05/2024-09/2024

- Architected the core geometry engine for the AccuContour system to deliver multi-organ 3D segmentation with submillimeter boundary precision that serves as the strictly constrained input for safety critical radiation dose planning in daily clinical operations, reducing manual contouring workload by 35%.
- Overcame physical reality gaps including breathing motion and metal artifacts by implementing a deformable registration module with anatomical regularization that maintained topology consistency and prevented mesh shearing across 1000 variable quality scans by modeling sliding interfaces between tissues.
- Built explainable, clinically oriented tooling for the segmentation pipeline, including contour-stability constraints on volume and centroid location that demonstrated a 28% reduction in shape drift when benchmarked against AccuContour's validated models using Dice and HD95. Additionally, integrated a retrieval-augmented VLM module that surfaces patient-specific risk factors and supporting oncology evidence for adaptive planning and QA.

EDUCATION

University of Illinois Chicago – *College of Engineering* 08/2025-09/2028

- *Ph.D. Student in Computer Science* Advisor: Philip S. Yu

Columbia University - *Fu Foundation School of Engineering* 09/2023-03/2025

- *Master of Science, Applied Mathematics* Advisor: Michael Tippett

University of North Carolina Wilmington - *College of Arts and Sciences* 09/2019-05/2023

- *Bachelor of Science, Mathematics* (Joint Dual Degree Program with Chongqing University of Arts and Sciences)

PUBLICATIONS

- **Chengze Li**, Wei Zhang, …, Philip Yu. "EndoTRAIL-RL: Trajectory-Supervised Agentic Reinforcement Learning for Multimodal 3D Gastrointestinal Endoscopy". *(Submitted to AAAI 2026)*
- **Chengze Li**, Lingwei Wei, …, Enze Ma, Philip Yu. "ARC-STAR: Auditable Post-Hoc Correction for PDE Foundation Models". https://arxiv.org/pdf/2605.22222. *(Submitted to NeurIPS 2026)*
- **Chengze Li**, Xiao Liu, …, Qichao Zhou, Philip Yu. "A Deployment Audit of Release-Side Risk in Conformal Triage under Prevalence Shift". *Conformal and Probabilistic Prediction with Applications (COPA) (Oral)*. 2026.
- Lingwei Wei, Dou Hu, **Chengze Li**, Wei Zhou, Songlin Hu, Philip Yu. "Geometry-aware Test-Time Adaptation on Graphs". *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. 2026.
- Hanrong Zhang, Shicheng Fan, …, **Chengze Li**, Wei-Chieh Huang, … Philip Yu. "CoEvoSkills: Self-Evolving Agent Skills via Co-Evolutionary Verification". *Conference on Language Modeling (COLM)*. 2026.